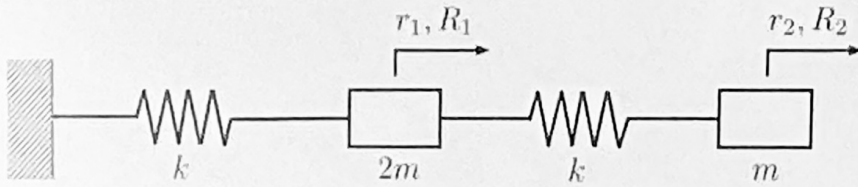


1. Find the determinant that, if equated to zero, leads to the natural vibration frequencies for:



2. Given the dynamic equilibrium below:

- Horizontal forces: $\sum F_x = 0 \Rightarrow 4m\ddot{r} + 4kr = 0 \Rightarrow m\ddot{r} + kr = 0$
- Moments: $\sum M_z = 0 \Rightarrow 2m\ddot{\theta}a^2 + 2k\theta a^2 = 0 \Rightarrow m\ddot{\theta} + k\theta = 0$

Find the mode shape associated with the natural frequencies.

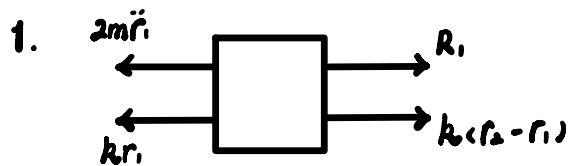
3. Demonstrate that the eigenvectors transform both the mass and the stiffness matrices of the system into a diagonal form with information below:

The equations of motion:

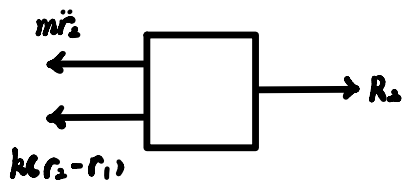
$$m \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \ddot{r}_1 \\ \ddot{r}_2 \end{bmatrix} + k \begin{bmatrix} 48 & -4 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$$

The matrix of eigenvectors is

$$\Phi = \begin{bmatrix} 1 & 1 \\ 4 & -4 \end{bmatrix}$$



$$2m\ddot{r}_1 + 2kr_1 - kr_2 = R_1$$



$$m\ddot{r}_2 + kr_2 = R_2$$

$$\therefore \begin{bmatrix} 2m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{r}_1 \\ \ddot{r}_2 \end{bmatrix} + \begin{bmatrix} 2k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$$

if $R = 0$, $r = r_0 \cos(\omega t)$

$$\begin{bmatrix} -2m\omega^2 + 2k & -k \\ -k & -m\omega^2 + k \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\therefore \det \begin{bmatrix} -2m\omega^2 + 2k & -k \\ -k & -m\omega^2 + k \end{bmatrix} = 0$$

$$\therefore (-2m\omega^2 + 2k)(-m\omega^2 + k) - k^2 = 2(m\omega^2)^2 - 4k(m\omega^2) + k^2 = 0$$

$$\therefore m\omega^2 = \frac{4k \pm \sqrt{8k^2}}{4} = (1 \pm \frac{\sqrt{2}}{2})k$$

$$\therefore \omega^2 = \frac{k}{m} \pm \frac{\sqrt{2}k}{2m}$$

$$\text{for } \omega^2 = (1 + \frac{\sqrt{2}}{2})\frac{k}{m}, \begin{bmatrix} (2 - \sqrt{2})k - 2 & -k \\ -k & (1 - \frac{\sqrt{2}}{2})k - 1 \end{bmatrix} \psi_1 = 0$$

$$\text{for } \omega^2 = (1 - \frac{\sqrt{2}}{2})\frac{k}{m}, \begin{bmatrix} (2 + \sqrt{2})k - 2 & -k \\ -k & (1 + \frac{\sqrt{2}}{2})k - 1 \end{bmatrix} \psi_2 = 0$$

60%

$$2. \begin{bmatrix} 4m & 0 \\ 0 & 2ma^2 \end{bmatrix} \begin{bmatrix} \ddot{r} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} 4k & 0 \\ 0 & 2ka^2 \end{bmatrix} \begin{bmatrix} r \\ \theta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$r = r_0 \cos(\omega t), \quad \theta = \theta_0 \cos(\omega t) \quad \therefore \ddot{r} = -\omega^2 r, \quad \ddot{\theta} = -\omega^2 \theta$$

$$\therefore \begin{bmatrix} -4m\omega^2 + 4k & 0 \\ 0 & -2ma^2\omega^2 + 2ka^2 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\therefore \det \begin{bmatrix} -4m\omega^2 + 4k & 0 \\ 0 & -2ma^2\omega^2 + 2ka^2 \end{bmatrix} = 0$$

$$\therefore (-4m\omega^2 + 4k)(-2ma^2\omega^2 + 2ka^2) = 8a^2(-m\omega^2 + k)^2 = 0$$

$$\therefore \omega = \pm \sqrt{\frac{k}{m}} \quad \therefore \Xi = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \text{ (as question 1)}$$

0%

For r ,

$$\omega_1^2 = \frac{k}{m}, \quad \psi_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{matrix} r \\ \theta \end{matrix}$$

For θ ,

$$\omega_2^2 = \frac{k}{m}, \quad \psi_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{matrix} r \\ \theta \end{matrix}$$

3. $\Phi^T = \begin{bmatrix} 1 & 4 \\ 1 & -4 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$ → normalized $\Phi = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$

$$\therefore m = \Phi^T M \Phi = \begin{bmatrix} 1 & 4 \\ 1 & -4 \end{bmatrix} m \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 4 & -4 \end{bmatrix}$$

$$= m \begin{bmatrix} 16 & 4 \\ 16 & -4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 4 & -4 \end{bmatrix} = m \begin{bmatrix} 32 & 0 \\ 0 & 32 \end{bmatrix}$$

$$k = \Phi^T K \Phi = \begin{bmatrix} 1 & 4 \\ 1 & -4 \end{bmatrix} k \begin{bmatrix} 48 & -4 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 4 & -4 \end{bmatrix}$$

$$= k \begin{bmatrix} 32 & 8 \\ 64 & -16 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 4 & -4 \end{bmatrix} = k \begin{bmatrix} 64 & 0 \\ 0 & 128 \end{bmatrix}$$

$$R = \Phi^T Q = \begin{bmatrix} 1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix}$$

$$\therefore m \begin{bmatrix} 32 & 0 \\ 0 & 32 \end{bmatrix} \ddot{q} + k \begin{bmatrix} 64 & 0 \\ 0 & 128 \end{bmatrix} q = \begin{bmatrix} 1 & 4 \\ 1 & -4 \end{bmatrix} Q$$

10%

